

VARUN BHADURGATTE NAGARAJ

Austin, Texas | (512)-212-6269 | varunnagaraj2009@gmail.com

EDUCATION

ROUND ROCK HIGH SCHOOL, ROUND ROCK, TX

EXPECTED GRADUATION: MAY 2027

Academy: Science, Technology,
Engineering and Mathematics (STEM)
and Business Industry Academy

Relevant Coursework: Advanced
Computer Science I, AP Precalculus,
Introduction to Engineering Design,
Principles of Engineering, AP Seminar,
TAG Adv English II

Honors: Top 10%, 1400 PSAT 10,

SCHOOL CLUBS & ACTIVITIES

DECA – Competed in the HRM 24
Emotional Intelligence event; gained
experience in business strategy, presentation,
and leadership under
timed case study conditions.

Mu Alpha Theta (Math Honor Society) –
Active member; volunteer math tutor
supporting underclassmen with algebra and
geometry concepts.

Cross Country (2023 Season) – Committed to
daily training and
improved 5K time from 35 minutes to 23
minutes (a 33% gain) through focused
conditioning and endurance work.

VEX V5 Robotics – Lead programmer
for competitive robotics team; developed
autonomous routines, custom PID
controllers, and real-time localization
systems to qualify for the VEX State
Championship.

SKILLS

Programming Languages: Python,

JavaScript, Java, C++, HTML, CSS

Tools & Frameworks: Flask, Wix Velo,
Requests, BeautifulSoup, JSON, Selenium,
JavaScript, HTML, CSS, Tailwind CSS,
Bootstrap, Supabase, GitHub, Render,
Python, YAML, Canva, Google Sites, Git
Bash.

Robotics Platforms: VEX V5 (PROS,
LemLib, VEXcode), FTC SDK (Java)

Languages Spoken: English, Kannada,
Hindi, Spanish

CERTIFICATIONS

BUSINESS INFORMATION MANAGEMENT

- Microsoft Office Specialist (Word,
Excel, PowerPoint) - May 2023

INTRO TO ENGINEERING DESIGN - 2024

- Autodesk Inventor Certified - April 2,
2024

BUSINESS MANAGEMENT

- NOCTI Business Management and
Administration - April 22, 2025

CTE INVOLVEMENT

ROBOTICS - VEX ROBOTICS

Engineer & Lead-Programmer | March 2023 – Present

- Built program using **Python** and later transitioned to **C++** to utilize enhanced motor control
- Developed a custom **PID controller** (5× consistency boost), **real-time localization** system through sensor integration (**90%+ autonomous success**), and exponential joystick scaling.
- Led programming using **PROS** and **LemLib**, demonstrating adaptability, technical leadership, and contributing to the team's **State Championship qualification**.

HALLHOP – DIGITAL PASS SYSTEM FOR SCHOOLS

Founder & Lead Developer | Jan 2025 – Present

- Built Chrome Extension with **Flask** based **REST API** to replace paper passes, enabling **real-time tracking** to reduce class **skipping** and **vaping**; implemented dynamic schedule detection (A/B days, lunch periods), and UI animations.
- Developed and optimized a **HAC data scraping API** by replacing **Selenium** with faster **Requests** and **BeautifulSoup**, significantly improving performance and **reducing** loading times by over **7x**.
- Integrated **Supabase** for automated logging, duration tracking, multi-student management, and session recovery; ensured scalability and security through a **CORS-enabled**, cloud-deployable API architecture.
- Currently prototyping a real-time indoor positioning system using **Aruba's Location Scanning API**, aiming for **2–4 meter accuracy** to support emergency response and live student mapping through Wi-Fi infrastructure during **school shootings** or other **critical events**

ROBOTICS - FIRST TECH CHALLENGE

Team Member | Jan 2025 – Present

- Programmed **autonomous** and **teleop** routines in Java using the **FTC SDK (LinearOpMode, OpMode, Telemetry, HardwareMap)**; developed feed-back functions & autonomous navigation logic.
- Integrated **AprilTag** computer vision for marker-based decision-making; gaining real-world experience in **Java** and **FTC SDK libraries**.
- Contributed to drivetrain **assembly**, **wiring**, and **hardware testing** to ensure full system readiness for matches.

ROUND ROCK HS CO-OP STORE – E-COMMERCE WEBSITE

Lightspeed/IT Team | Jan 2025 – Present

- Re-photographed, reformatted, and rewrote listings for **400+** items; verified inventory for catalog accuracy.
- Integrated **Stripe** to enable **Google Pay** and **Apple Pay** at checkout; improved UX with **interactive CSS animations**.
- Collaborated with the principal, district CTEs, and staff to **streamline operations**, **reduce class-skipping** through delivery-based ordering, and expand roles for FAC and **disabled students** via **inclusive** fulfillment.

VOLUNTEERING

Boy Scouts of America | Life Scout | 2023 – Present

Led younger scouts at summer camp, managing **safety**, **scheduling**, and **mentorship**. Held troop leadership roles focused on **conflict resolution** and **morale**; currently two months from achieving Eagle Scout rank.

Coding & Career Opportunities for Orphans | Co-Founder | Nov 2024 – Present

Launched an initiative to teach **Python** and career skills to **orphaned youth** in India with limited tech access. Partnered with a local orphanage and **Steel City Codes** to adapt curriculum; working with **Hack Club Bank** to obtain **501(c)(3)** status.

Austin Kannada Sangha (AKS) | Event & Technical Volunteer | 2020 – Present

Supported **AV setup**, mic testing, speaker coordination, and **online ticketing** for events with **1,000+** attendees. Helped manage guest **check-in**, **food service**, and **backstage logistics**; **emceed** a live cultural program.

ATX2 Project – Meals for Students | SEWA LEAD Volunteer | 2024 – Present

Researched **food insecurity** and partnered with **Texas Baptist Children's Home**. Launched a **Google Sites** platform connecting students to food programs; designed **outreach** using Canva and social media.